```python
a = 0
b = 0

while a != 2:
    if a == 0:
        a = 4
    elif b == 3:
        b = 0
    else:
        x = min(a, 3 - b)
        a = a - x
        b = b + x

    print("4L Jug =", a, "  3L Jug =", b)
```

```
4L Jug = 4   3L Jug = 0
4L Jug = 1   3L Jug = 3
4L Jug = 1   3L Jug = 0
4L Jug = 0   3L Jug = 1
4L Jug = 4   3L Jug = 1
4L Jug = 2   3L Jug = 3

Process finished with exit code 0
```